2nd OpenFOAM HPC Challenge (OHC-2): Hardware Track Summary

JULY 1-2, 2026, Guimarães

Mini-symposium to be held in the auspices of the
2026 OpenFOAM Workshop

Organized by the OpenFOAM HPC Technical Committee (TC)

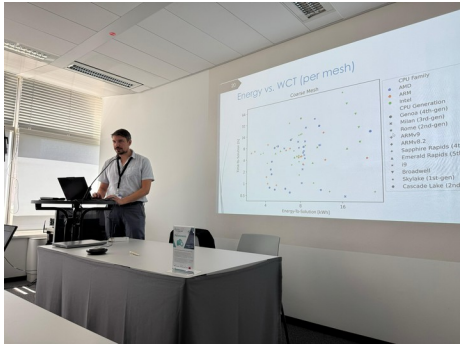
Presented by:
Sergey Lesnik, Gregor Olenik, Mark Wasserman

OpenFOAM HPC Challenge (OHC)

Background

- **The OpenFOAM HPC challenge is an annual event that runs in conjunction with the OpenFOAM workshop**
- **The event brings together hardware vendors, industrial users and developers to present recent advances in optimizing OpenFOAM performance**
- **Participants submit simulation results obtained**
  - on **state-of-the-art hardware - Hardware Track**
  - with various **software enhancements - Software Track**
- **Results are analyzed, discussed and disseminated to the community:**
  https://arxiv.org/abs/2603.27565

  github.com/OpenFOAM-HPC-Challenge





2nd OpenFOAM HPC Challenge (OHC-2)

Objectives

- **Evaluate the efficiency and scalability of OpenFOAM on different hardware platforms (hardware track)**
- **Compare the performance of latest OpenFOAM code enhancements and flavors (software track)**
- **Propose unified metrics that will help guide the future development of OpenFOAM on next-generation hardware**
- **Bring together hardware vendors, industrial users and developers to discuss OpenFOAM performance**
- **Publish benchmark performance results of OpenFOAM on state-of-the-art hardware for guidance to users and developers**

2nd OpenFOAM HPC Challenge (OHC-2)

Scope

- **The focus of this mini-symposium will be the industrial testcase of external flow over a static DrivAer automotive model, investigated as case 2 in the AutoCFD2 workshop, and as case 2a in the subsequent AutoCFD3 & AutoCFD4 workshops**
- **A full OpenFOAM setup* of the testcase, including meshes, physical model, and solver selection can be found here**
- **Reference numerical results shall also be made available soon**
- **In the hardware track, participants are invited to submit and present results obtained with the pre-defined setup using an official OpenFOAM version, on any combination of hardware**
- **In the software track, participants are invited to submit and present results obtained with any OpenFOAM version and custom solver, on any hardware, so long as the mesh and physical modelling remains true to source, and accuracy is retained**
- **Participants are also invited to submit and present HPC enhancements related to pre- and post-processing of OpenFOAM data, applied to the above testcase**

*Special thanks to Upstream CFD for providing the OpenFOAM DrivAer setup, which has been adapted for the challenge



Agenda

July 1

11:00-11:20	Introduction + Keynote lecture (Ivan Spisso)
11:20-11:40	Summary of Hardware Track Contributions (Sergey Lesnik)
11:40-11:50	Hardware Track Participant Talks (Eike Tangermann)
11:50-12:00	Hardware Track Participant Talks (Lydia Schulze)
12:00-12:10	Hardware Track Participant Talks (Flavio Galeazzo)
12:10-12:20	Hardware Track Participant Talks (Mark Wasserman)
12:20-12:40	Summary of Software Track Contributions(Mark Wasserman)
12:40-15:00	*Lunch break + Planery*
15:30-15:50	Extension (optional) of hardware track

July 2

11:00-11:20	Software Track Participant Talks 1 (Simone Bnà)
11:20-11:40	Software Track Participant Talks 2 (Stefano Oliani)
11:40-12:00	Software Track Participant Talks 3 (Gregor Olenik)
12:00-12:20	Software Track Participant Talks 4 (Henrik Rusche)
12:20-12:40	Software Track Participant Talks 5 (Flavio Galeazzo)
12:40-15:00	*Lunch break + Planery*
15:30-15:50	Summary + Open Discussion

*** All times are local, GMT+1**

Contributors

- **148 data points contributed by 8 contributors:**
  - Wikki GmbH
  - Cineca
  - Universität der Bundeswehr München
  - Federal Waterways Engineering and Research Institute
  - Technical University Munich
  - Engys
  - HLRS
  - Nvidia

Thank you!

Breakdown of Contributions (Statistics)

- **Total number of valid entries: 148 (OHC1: 236)**
  - Hardware track entries 83, Software track entries 58
  - AMD 73 (106), Intel 0(80), ARM 10 (50)
- 30? different CPU models and 5 (3) GPU models

- Max nodes used: 512 (256)
- Max cores used: 65536 (32768)
- Min wall-clock time to solution: 5.5 (7.8) minutes
- Max wall-clock time to solution: 129 (65.7) hours
- Min energy of single simulation: 1.1 (2.1) [kWh]
- Max energy of single simulation: 103 (236.9) [kWh]
- Max performance: 67.7 [MfvOps/W]

Breakdown of Contributions (Statistics)

- Total reported energy consumption:

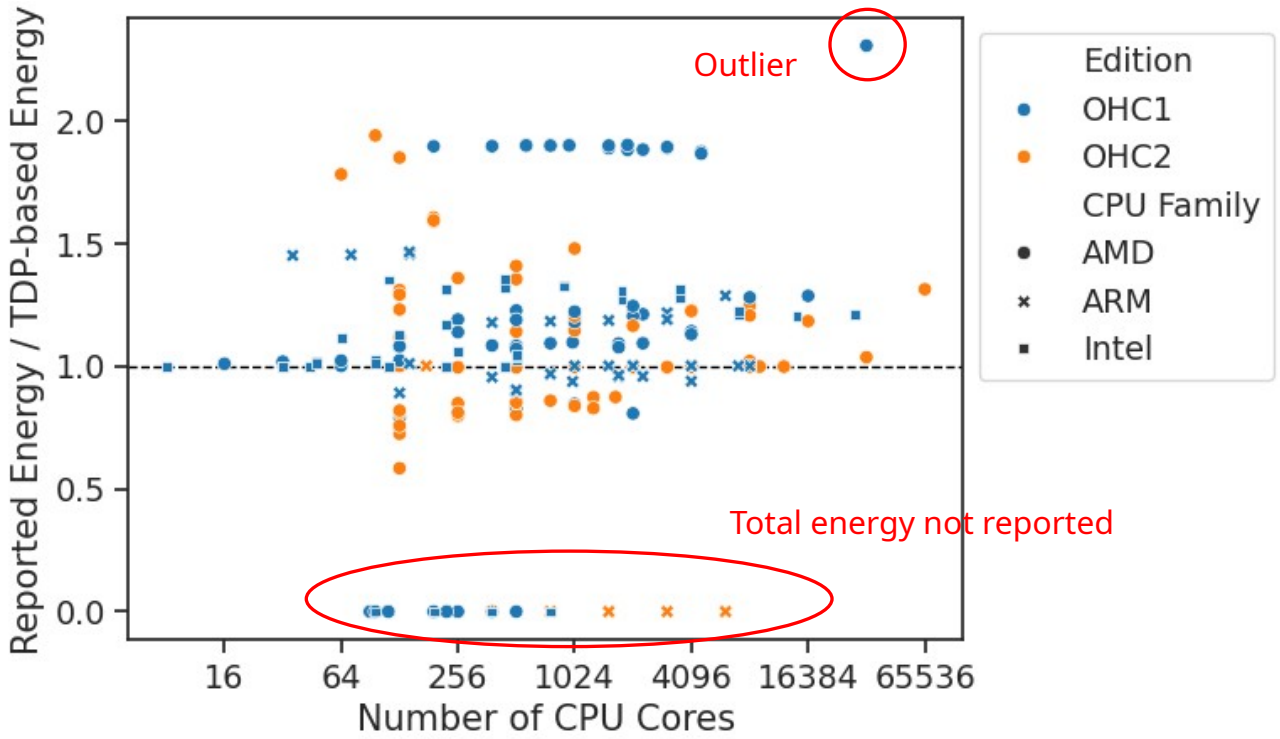
  2152 kWh, potentially bringing ~100 bath tubs to a boiling point!



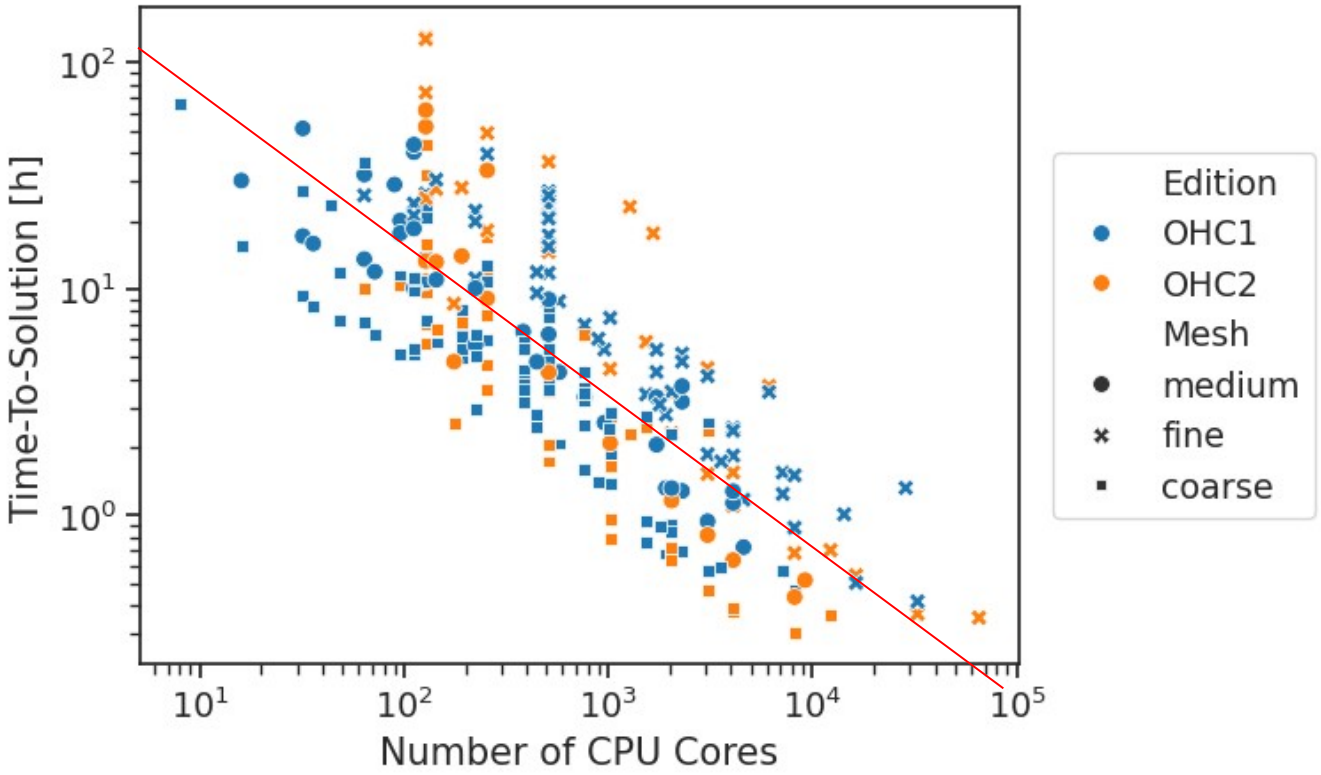
Issues, Assumptions and Disclaimers

- **No. of cores was corrected to represent physical cores (hyper-threading disregarded)**
- **Changed names/values to adhere to a predefined scheme for easier evaluation**
- **The focus lied on providing an overview of the complete data set, not analysis of single simulations or submissions - the latter are covered by the participants' talks**
- **Added some values based based on publicly available data (e.g. TDP according to the CPU model)**
- **All energy reports are based on TDP values**

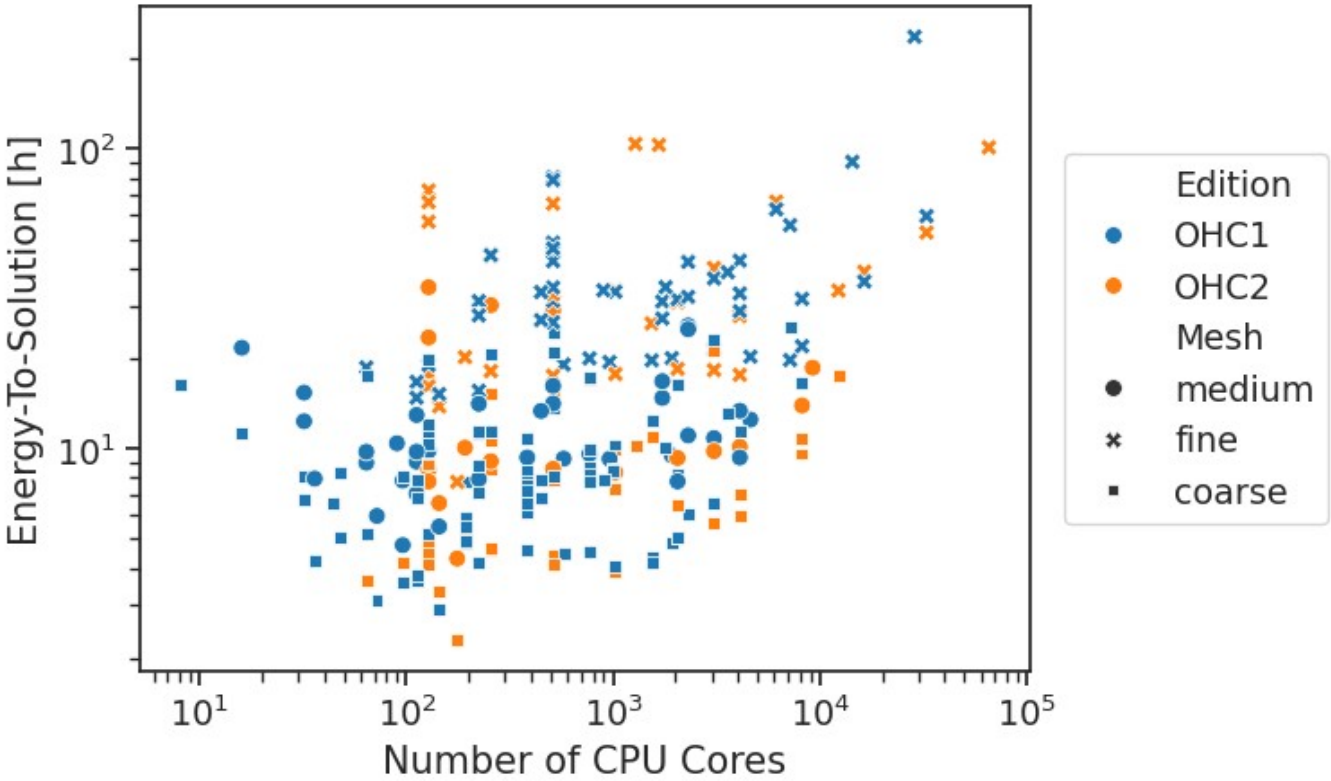
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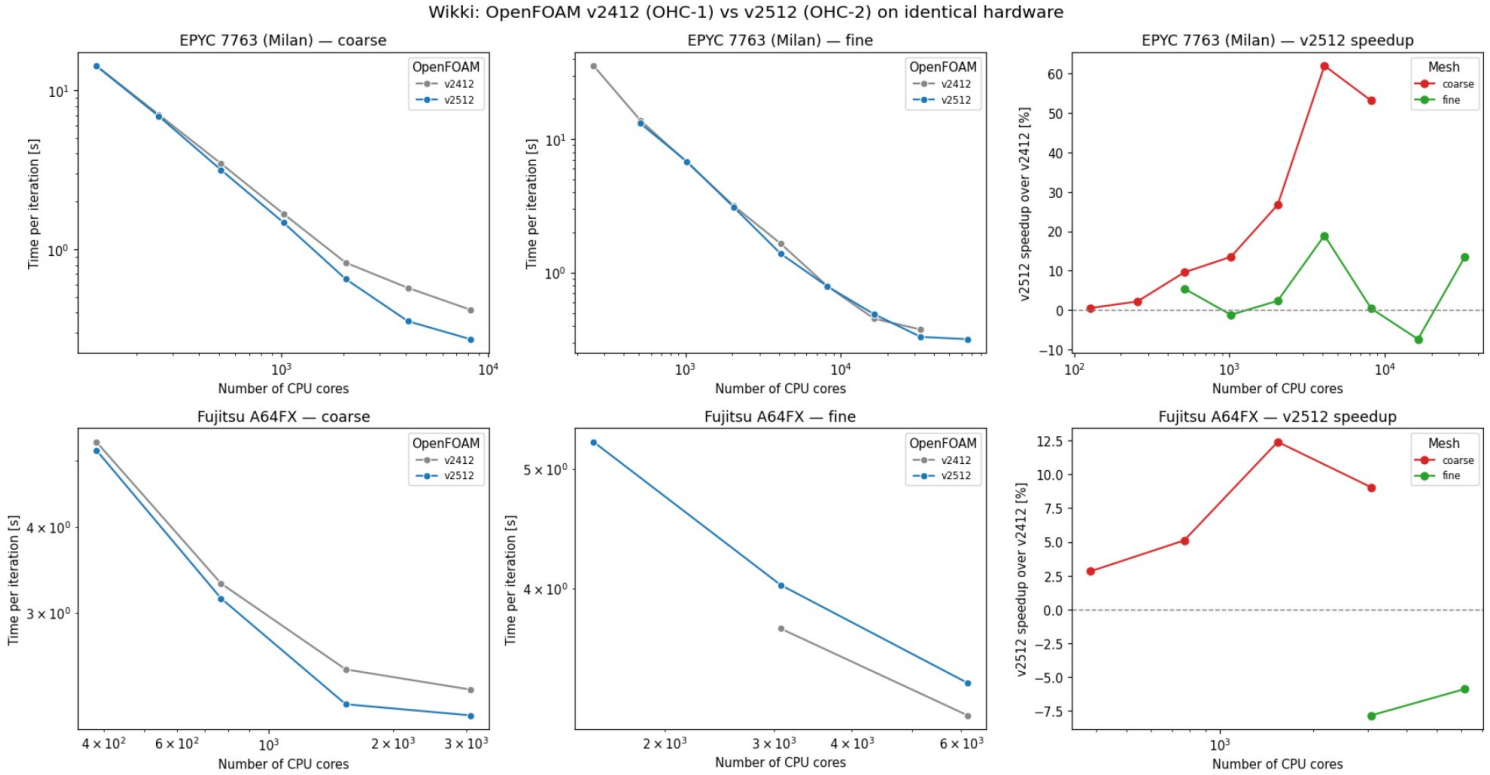
Breakdown of Contributions TTS



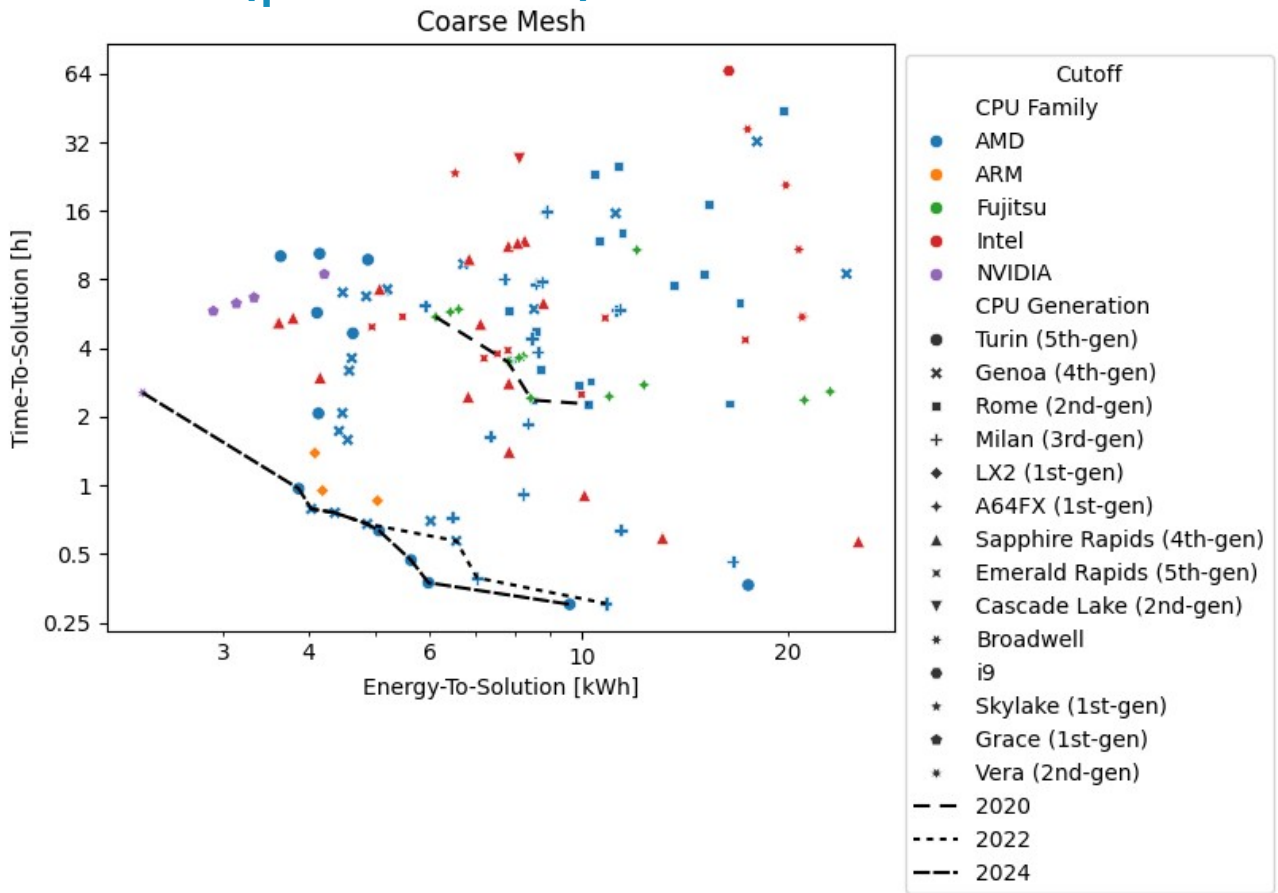
Breakdown of Contributions ETS



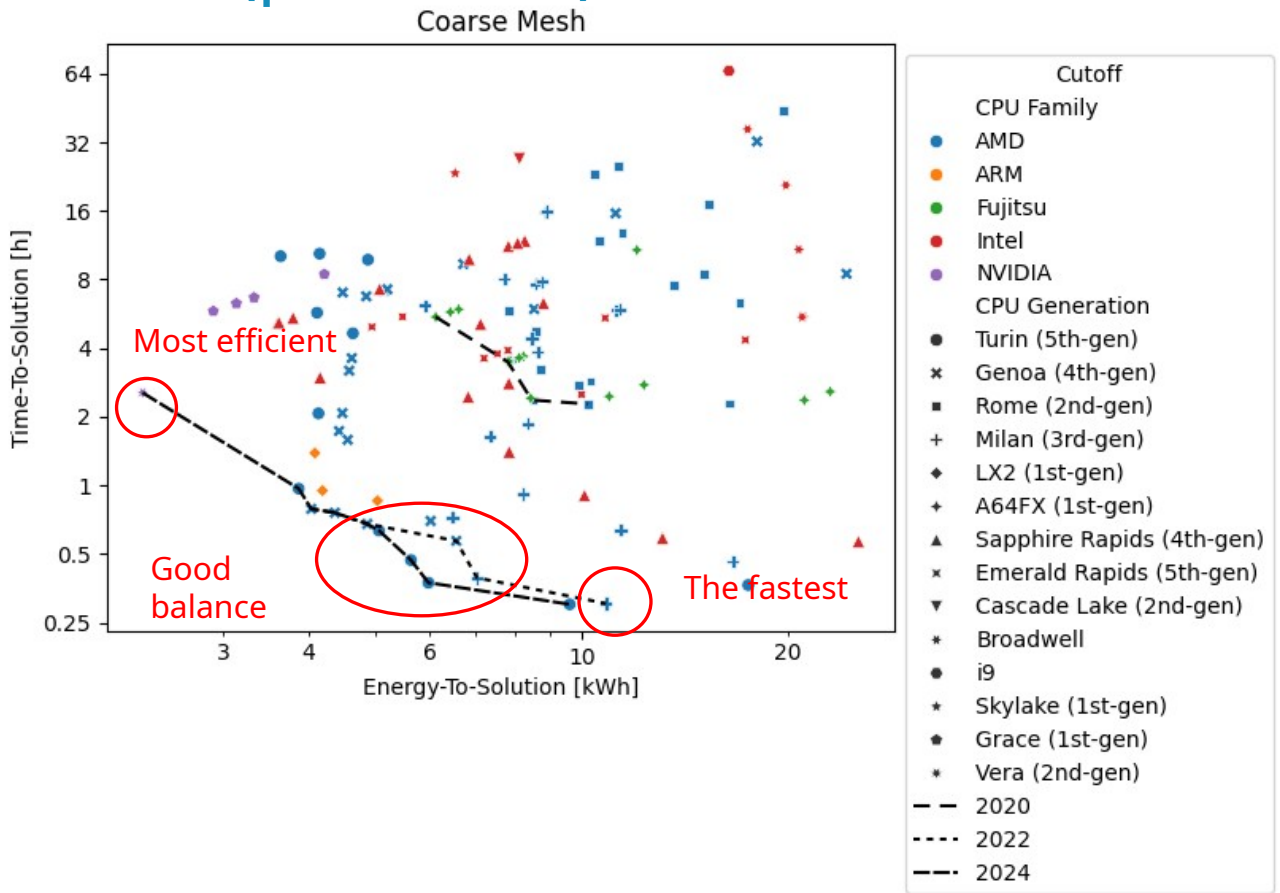
OpenFOAM v2412 vs. v2512



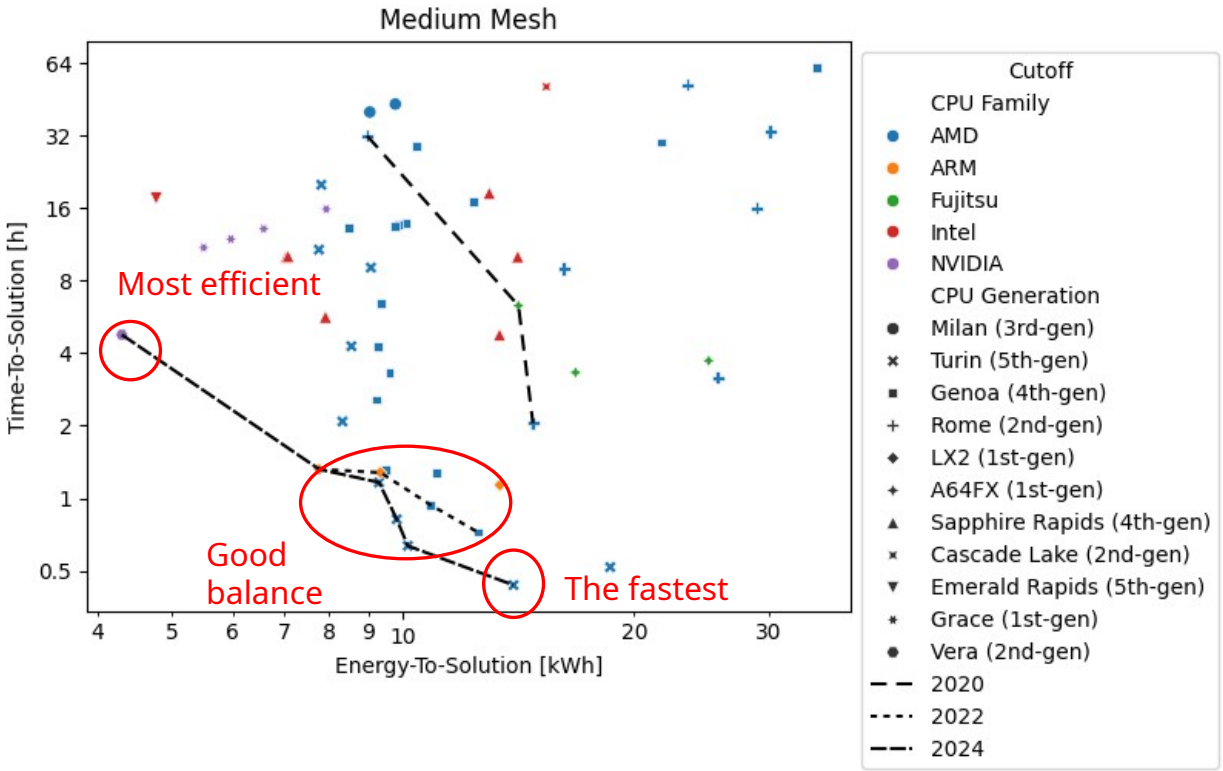
ETS vs. TTS (per mesh)



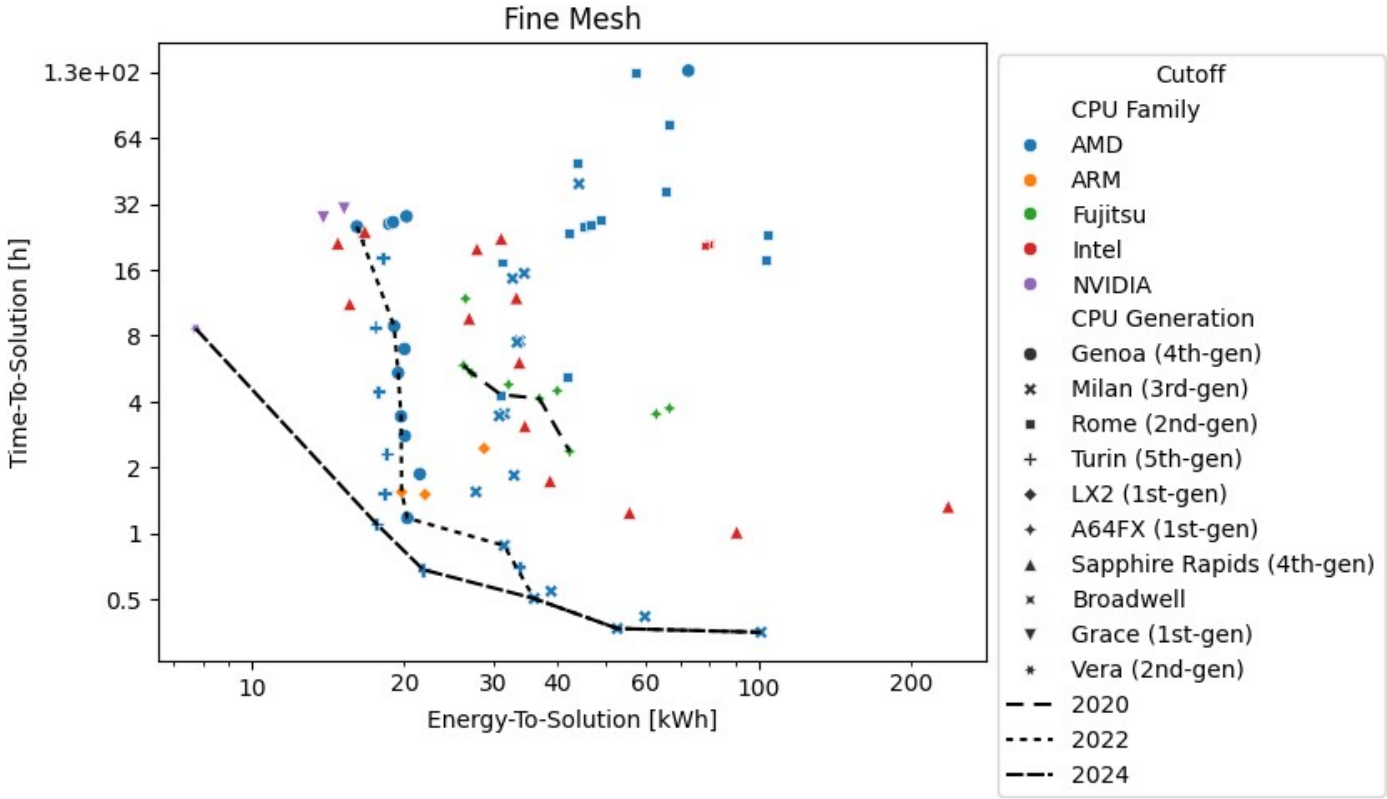
ETS vs. TTS (per mesh)



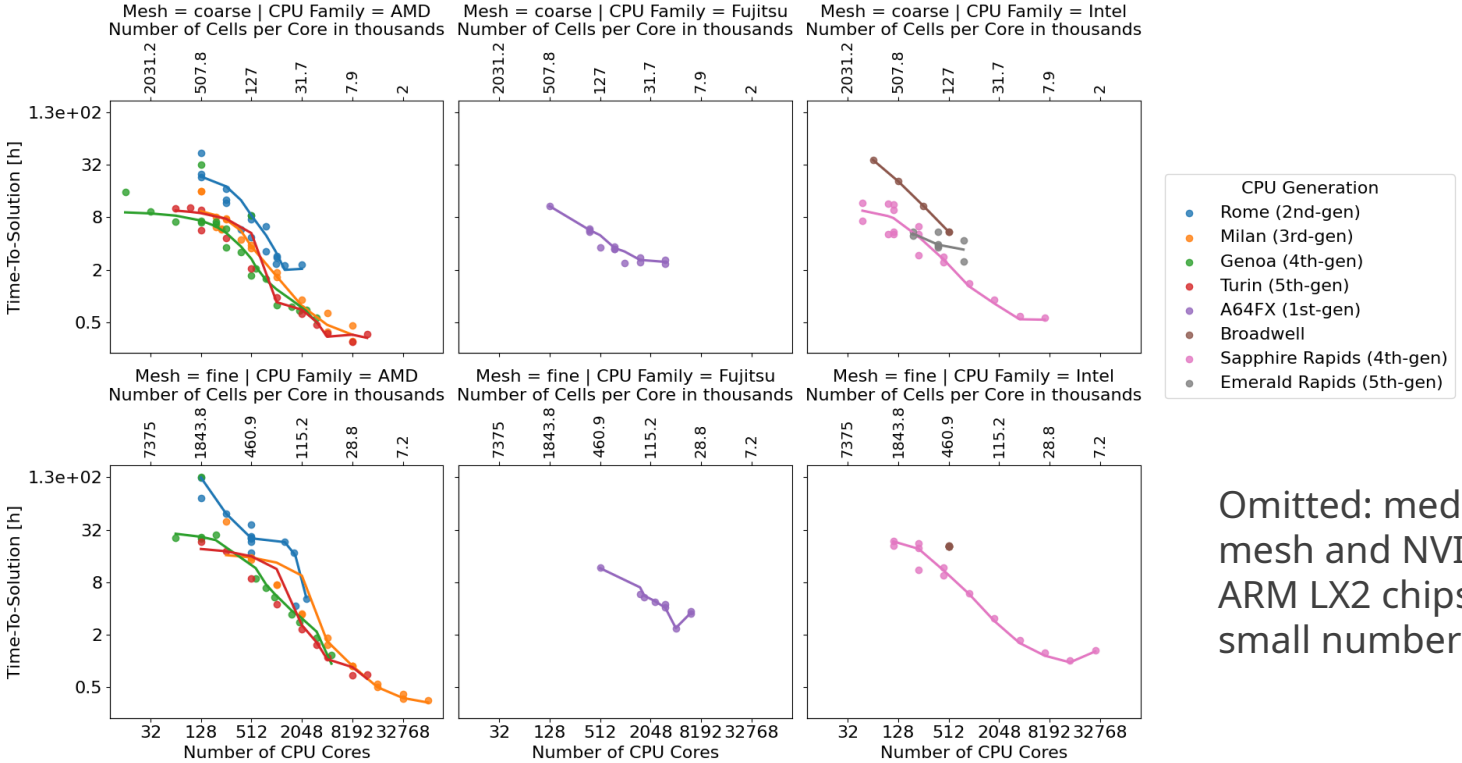
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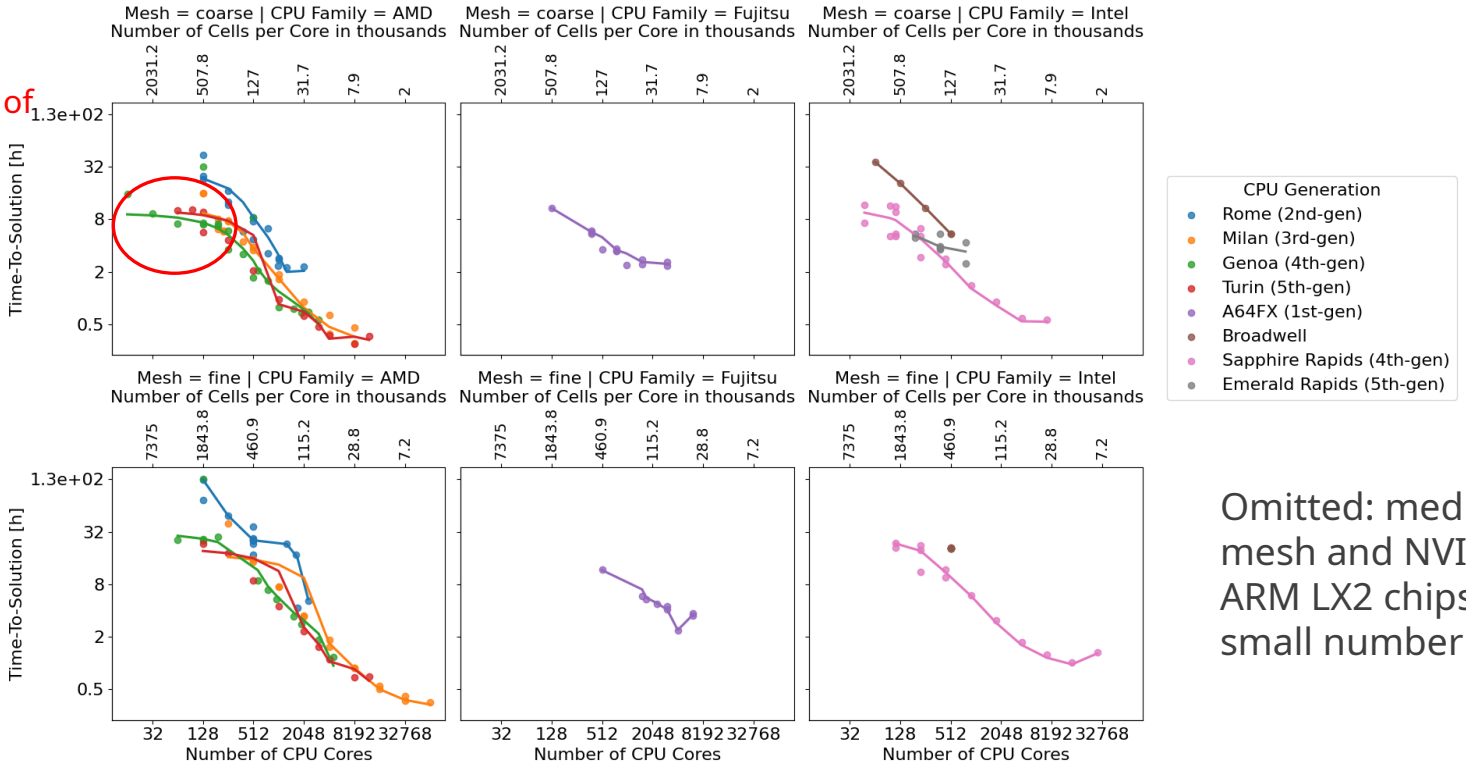
Scalability Limits (Strong Scaling)



Omitted: medium size mesh and NVIDIA + ARM LX2 chips due to small number of points

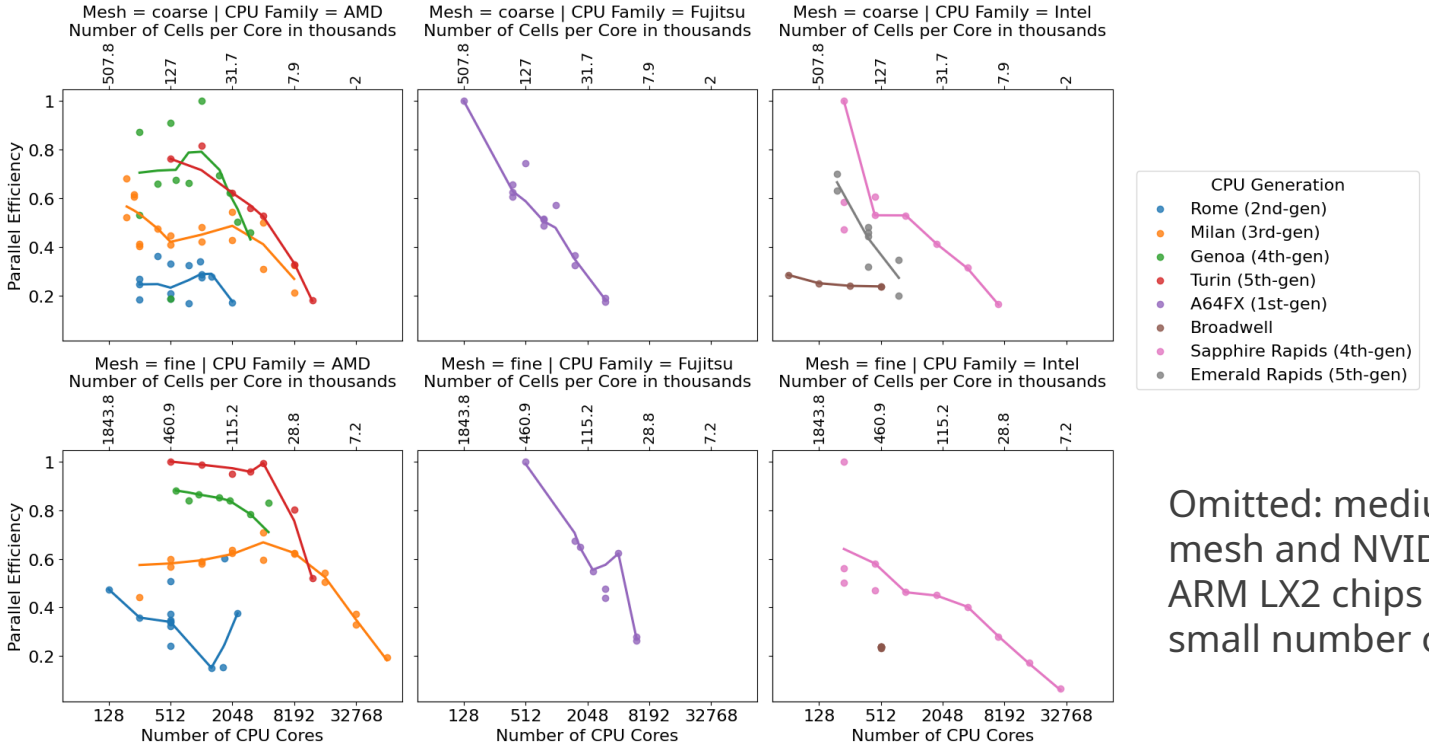
Scalability Limits (Strong Scaling)

Scaling limit of single-node (mem. BW)



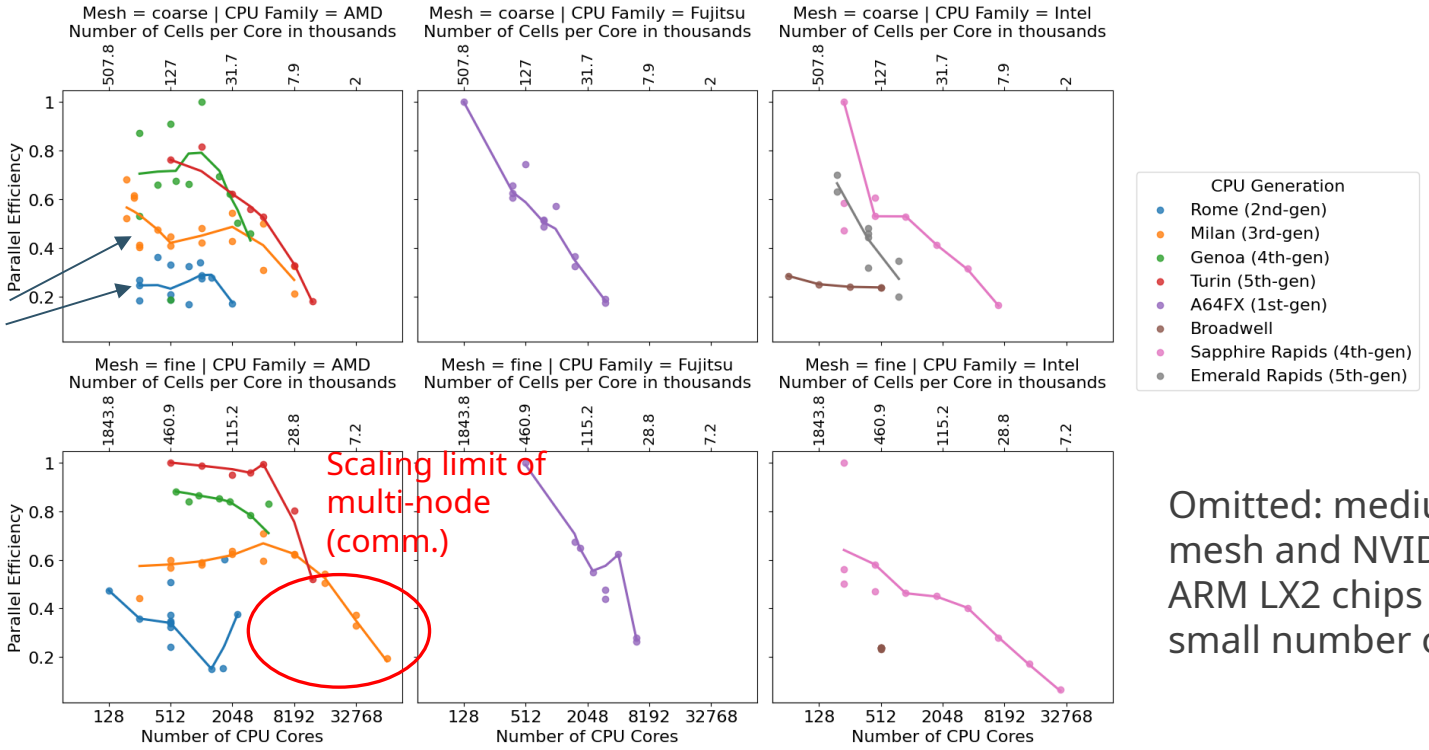
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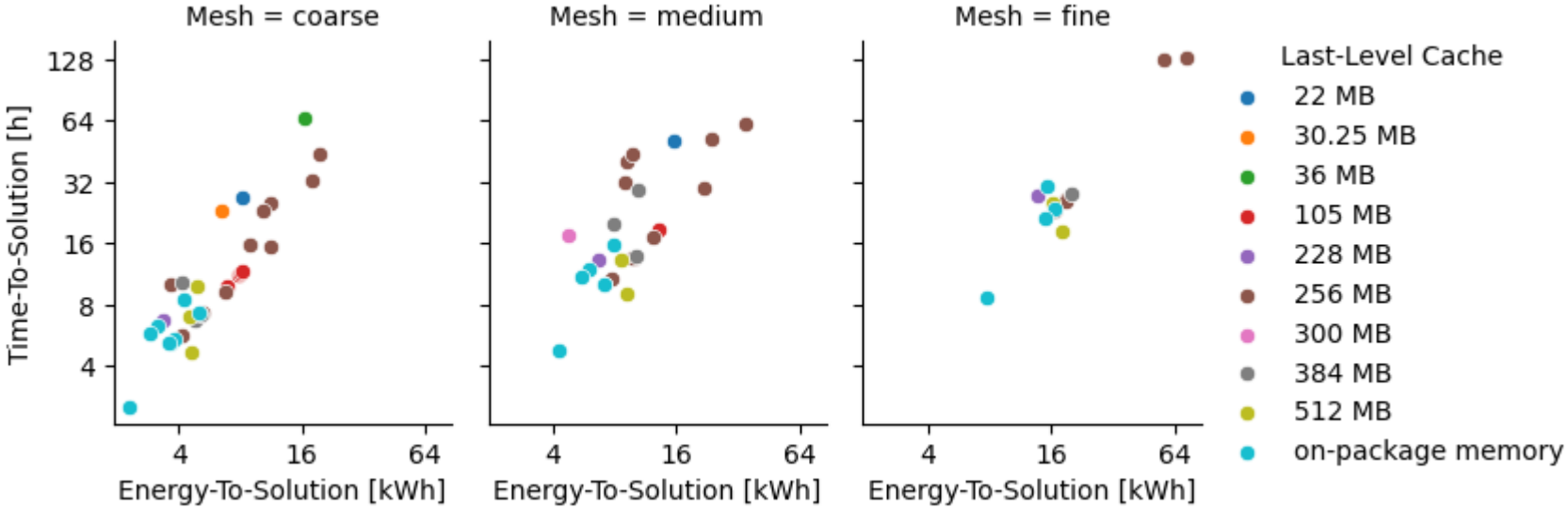


Offset between generations

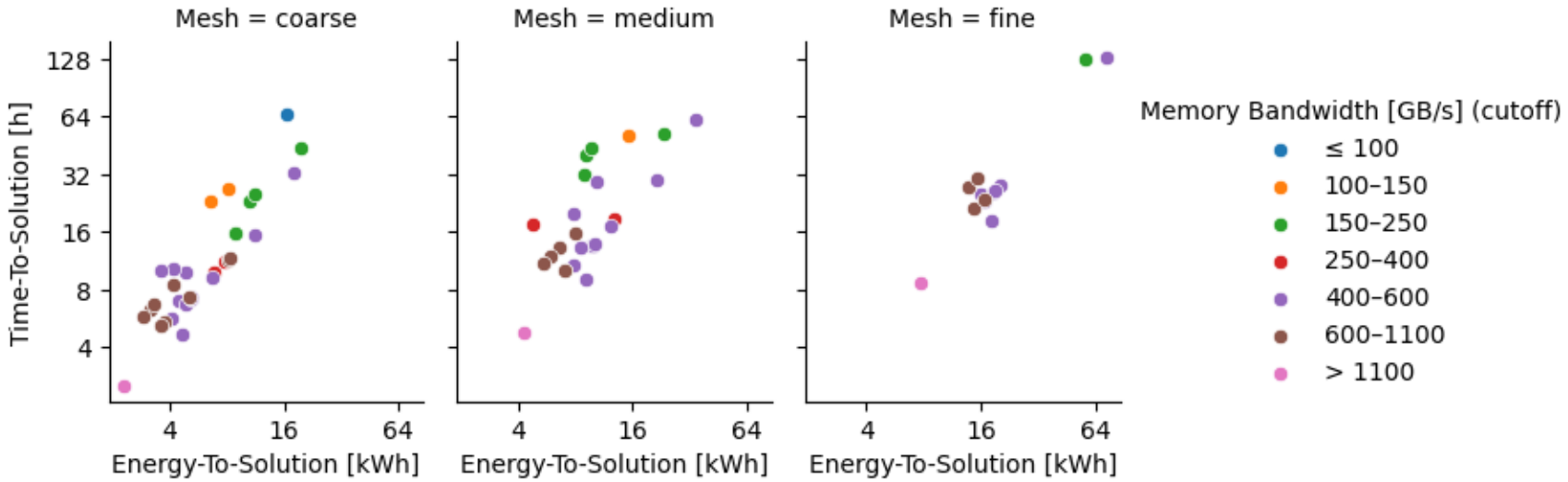
Scaling limit of multi-node (comm.)

Omitted: medium size mesh and NVIDIA + ARM LX2 chips due to small number of points

Single Node: LLC/On-Package Memory



Single Node: Memory Bandwidth



Summary

- **Contributions were down this year, need to refresh the topic for the next challenge**
- **GPU-accelerated OpenFOAM redefines energy efficiency**
  - Best GPU run at 2.3× the best CPU and ~3.5× the median CPU
- **For CPU-based OpenFOAM, expansion of fast on-package memory (LPDDR5X/HBM) increases performance by 2x compared to previous generation**
- **Identified pareto front of optimal balance between time-to-solution and energy-efficiency**
  - Accelerators own the low-energy end (greenest solutions), while massive CPU scale-out owns the fast end (lowest time-to-solution, at a steep energy cost)
  - No single approach dominates both axes
- **Multi-node strong scaling still limited at the <10k cell/core level**
- **All data will be available on our repo:**
  - https://github.com/OpenFOAM-HPC-Challenge/
- **Analysis to appear in OFJ (under review)**